

The empirical recurrence for OEIS A238307: recovering a conjecture that is not written down

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Abstract

OEIS A238307 records “Empirical recurrence of order 76”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 1776$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 76, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 76 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A238307 is “Number of $(n+1) \times (2+1)$ 0..3 arrays with the maximum minus the lower median of every 2×2 subblock differing from its horizontal and vertical neighbors by exactly one.”. It has offset 1 and begins

1776, 20128, 227622, 2570594, 29039618, 328887234, 3724943100, 42156330796, ...

The entry states:

Empirical recurrence of order 76 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 10:50:53, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 1776$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 1776$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 3552$ exact terms of the model returns order 76 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{76} c_i a(n-i),$$

c_1	0	c_{20}	143819948233	c_{39}	1098815982455005785	c_{58}	2552131844208767656
c_2	0	c_{21}	−949452193524	c_{40}	1878496926360289268	c_{59}	−1985392340164925609
c_3	590	c_{22}	−3466310431313	c_{41}	−4154137393506684650	c_{60}	−308502195280952876
c_4	6047	c_{23}	20506201952776	c_{42}	−2489534202633012199	c_{61}	2002321901889236764
c_5	33282	c_{24}	−8028535411851	c_{43}	5512403342728275355	c_{62}	344130666354013184
c_6	143172	c_{25}	−375549140588226	c_{44}	606241215183684002	c_{63}	−262987165354583778
c_7	−51847	c_{26}	−67909347802678	c_{45}	−1593510698063116450	c_{64}	271094423939197184
c_8	−3383143	c_{27}	1672222434601647	c_{46}	3736090968544920876	c_{65}	229978471954441064
c_9	−19098221	c_{28}	−1014167853197958	c_{47}	−4841764212703579592	c_{66}	−55304525405563700
c_{10}	−62660439	c_{29}	−3819901933221444	c_{48}	−6977069500653741268	c_{67}	−141196591281205736
c_{11}	50129967	c_{30}	12177171250980535	c_{49}	5774208940465613649	c_{68}	−16903093896289976
c_{12}	1167166401	c_{31}	11379879333755080	c_{50}	5855695151473940574	c_{69}	22788803756890560
c_{13}	3926502816	c_{32}	−19703330851331473	c_{51}	−962200406677614129	c_{70}	11349407149777184
c_{14}	4633770290	c_{33}	−5046728126887472	c_{52}	−3635673360797615615	c_{71}	139234319179520
c_{15}	−30502410747	c_{34}	−107938157967819317	c_{53}	−2689068586524211097	c_{72}	−2069352319020288
c_{16}	−161347026851	c_{35}	−157569757784410886	c_{54}	3797403507458193497	c_{73}	−660849214850048
c_{17}	−92876223654	c_{36}	466875447950918420	c_{55}	2229508234425503651	c_{74}	−93674083704832
c_{18}	780089150443	c_{37}	254581489443163362	c_{56}	−3424368555467162215	c_{75}	−6774341369856
c_{19}	1245151657046	c_{38}	−971895742022921422	c_{57}	1184789416489364860	c_{76}	34665922560

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 76, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $76 + 4$ terms removed returns order 76 as well, so $\deg q_{\min} = 76 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 15 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $76 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 76 that this sequence satisfies — and that family turns out to have exactly one member.

References

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